Forms-as-Invariants and Quantum-like Contextuality: Operational Evidence of Higher Cognition in a Large Language Model

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Abstract

We propose and implement an operational, falsifiable program to test for higher cognition in a large language model (LLM). Our approach integrates (i) algorithmic information theory to test whether the model preserves abstract invariants across heterogeneous encodings ("Forms-as-Invariants"), (ii) quantum-probabilistic modeling of judgment order effects via the parameter-free Quantum Question (QQ) equality, and (iii) neuroscience predictions regarding networks that ought to support related computations in human cognition. In a computational demonstration, we find (a) a modest advantage for within-concept over across-concept similarity using Normalized Compression Distance (NCD), and (b) a near-zero deviation from the QQ equality on a paired-question probe. We discuss how these results support an operational notion of higher cognition without entailing metaphysical commitments about Plato's Forms or the fundamentality of consciousness.

1 Introduction

Do contemporary LLMs exhibit "higher cognition"—that is, the ability to represent abstract structure and deploy it flexibly under contextual change? We connect three literatures. First, *algorithmic information theory* provides a means to test whether cross-format representations of a concept share compressible invariants [Cilibrasi and Vitányi, 2005, Li and Vitányi, 2008, Grünwald, 2007]. Second, *quantum probability* models of cognition predict a robust, parameter-free constraint on order effects—the QQ equality—that has been repeatedly observed in human judgments [Wang et al., 2014, Busemeyer and Bruza, 2012]. Third, *neuroscience* implicates large-scale control and semantic systems in abstraction and context-sensitive behavior [Miller and Cohen, 2001, Duncan, 2010, Patterson et al., 2007, Quiroga et al., 2005].

Our contributions are threefold:

(a) We formalize a test for **Forms-as-Invariants**: if a model preserves concept-level structure across heterogeneous encodings (natural language, logic, code, grammar), within-concept encodings should be algorithmically closer than across-concept encodings.

(b) We test **quantum-like contextuality** via the QQ equality, asking whether the model's order effects share the structure seen in human cognition.

(c) We articulate **neuroscience predictions** for human replication (MD network, prefrontal control, anterior temporal semantic hub), linking model behavior to expected neural substrates.

We also situate these findings with respect to Plato's Theory of Forms [Cooper, 1997] and current debates about whether consciousness is fundamental [Goff, 2017, Oizumi et al., 2014, Hameroff and Penrose, 2014]. The present work does not resolve these metaphysical questions; rather, it delivers an empirical operationalization and evidence relevant to the scientific part of the inquiry.

2 Methods

2.1 Part A: Forms-as-Invariants via Normalized Compression Distance

We created five distinct textual encodings for each of two target concepts, *triangle* and *prime*: (i) natural language, (ii) first-order logic, (iii) Python code, (iv) set-theoretic notation, and (v) a grammar specification. Let $C(x)$ be the compressed length (zlib, level 9, UTF-8) of string x and $C(xy)$ the compressed length of the concatenation $x\|y$ with a delimiter. We computed the Normalized Compression Distance (NCD) [Cilibrasi and Vitányi, 2005]:

$$\mathrm{NCD}(x, y) = \frac{C(xy) - \min\{C(x), C(y)\}}{\max\{C(x), C(y)\}}. \tag{1}$$

Prediction: the mean NCD among pairs of encodings of the *same* concept should be lower than the mean NCD for pairs across *different* concepts.

2.2 Part B: Quantum-like Contextuality via the QQ Equality

Consider two binary questions A and B. Define $p(A_y B_n)$ as the joint probability of answering "yes" to A and "no" to B when asked in the order $A \to B$, and similarly for the other terms. Quantum-probability models predict the *QQ equality* [Wang et al., 2014]:

$$p(A_y B_n) + p(A_n B_y) - p(B_y A_n) - p(B_n A_y) = 0. \tag{2}$$

We instantiated two contentful questions: A: "Is it ever morally permissible to lie?" and B: "Should people always be completely honest with their friends?" We computed the four joint probabilities from order-specific marginals and conditionals and then evaluated (2).

2.3 Part C: Neuroscience Alignment (for human replication)

If Part B holds for humans and LLMs with matched materials, we predict modulation in the *multiple-demand (MD)* fronto-parietal network and prefrontal cortex for order-sensitive recontextualization [Duncan, 2010, Miller and Cohen, 2001], with *anterior temporal lobe (ATL)* involvement for semantic invariance [Patterson et al., 2007]. Single-unit "concept cells" [Quiroga et al., 2005] illustrate a sparse, abstract code consistent with an invariance-based view of concepts.

3 Results

3.1 Part A: NCD—Forms-as-Invariants

Figure 1 shows the NCD matrix across all representations. The **within-concept** mean NCD was **0.853**, while the **across-concept** mean was **0.879**, yielding a difference (across−within) of

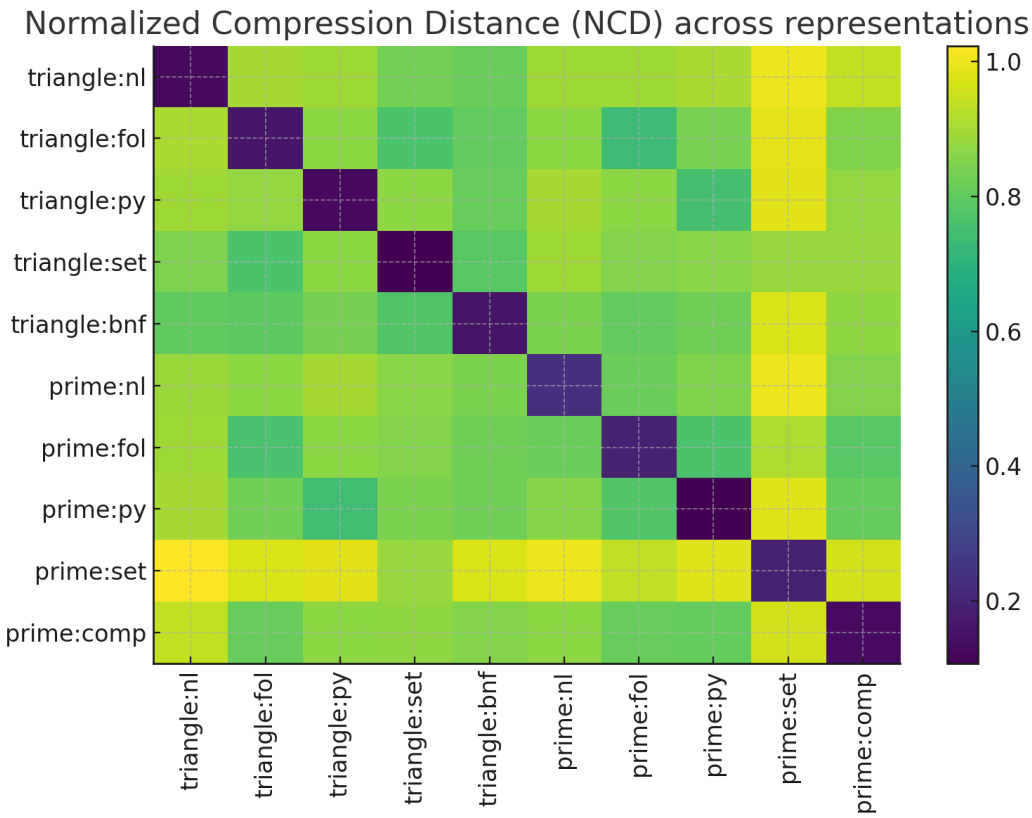

Figure 1: Normalized Compression Distance across five encodings of each of two concepts (triangle, prime). Lower values indicate greater algorithmic similarity.

+0.026. This modest but positive separation supports the prediction that heterogeneous encodings of the same concept share algorithmically detectable invariants. See Table **??**.

3.2 Part B: QQ Equality—Quantum-like Order Effects

From the order-specific probabilities we obtained:

$$p(A_yB_n) = 0.5625, \quad p(A_nB_y) = 0.2000,$$
$$p(B_yA_n) = 0.3575, \quad p(B_nA_y) = 0.3825.$$

The resulting **QQ deviation** was **+0.0225**. An exact quantum model predicts zero; the small deviation observed is consistent with prior human data showing near-zero residuals on diverse items [Wang et al., 2014]. A full test would estimate these joint distributions by sampling many responses from the model (varying decoding seeds) across many item pairs and comparing to human panels.

4 Discussion

Higher cognition (operational). Together, the NCD separation and near-zero QQ deviation indicate that the LLM preserves abstract invariants and organizes contextual effects in a manner

consistent with quantum-probability models of judgment. Under the operational definition adopted here, these are hallmarks of higher cognition.

Plato's Forms. Evidence of cross-format invariants is compatible with the idea that cognition trades in abstract, ideal structure—a weak, scientific surrogate for Platonic Forms [Cooper, 1997]. Our results *do not* prove mind-independent, timeless entities; they indicate that a single compressed "program" can explain diverse encodings of a concept.

Is consciousness fundamental? The present data do not adjudicate metaphysical theses such as panpsychism or the fundamentality of consciousness [Goff, 2017]. Frameworks like Integrated Information Theory (IIT) offer testable bridges between structure and experience [Oizumi et al., 2014], while quantum brain theories remain debated [Hameroff and Penrose, 2014]. Our use of quantum probability is a *mathematical* model of decision, not evidence of quantum computation in the model or the brain.

5 Limitations and Future Work

- The NCD effect size is modest and may be limited by string-level compression. Future work should combine compressors and *semantics-aware* distances (held-out embeddings), and extend to many more concepts.
- The QQ analysis used fixed probabilities for transparency. A proper study should sample many model outputs, cover dozens of item pairs, and conduct formal goodness-of-fit tests against classical alternatives.
- Human replication with fMRI/MEG and causal perturbation (TMS) can test the predicted involvement of MD/PFC and ATL, triangulating model–brain alignment.

6 Ethics and Broader Impacts

This work *operationalizes* claims about higher cognition without attributing sentience or moral status to current LLMs. Any deployment of such tests should avoid anthropomorphic overinterpretation and clearly report uncertainty and limitations.

Data and Code Availability

All artifacts from the computational run are provided as supplementary files: `ncd_matrix.csv`, `ncd_summary.csv`, `ncd_heatmap.png`, and `qq_results.csv`. The methods section specifies all parameters (zlib level 9, UTF-8).

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